LOF CRM – E-mail Integration Setup Flow

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Purpose

The CRM can send e-mail to parents today once Resend is set up, but it cannot receive e-mail and cannot send staff-written e-mails about a student. This document is the setup flow for both.

The scenario it covers:

- **Incoming:** parents and clients write to a Lab of Future address, such as `admissions@`, and the message appears in the CRM on the right lead or student.
- **Outgoing:** staff send e-mails about a student (progress, attendance, fee reminders) to the parent's e-mail address on record, and parent replies come back into the CRM.

Current status

Three of the six e-mail capabilities work today; two need new development and one needs a small fix.

Capability	Status	Notes
E-mail to a lead's parent (enquiry stage)	Works	Sent through Resend once configured; blocked if the lead has no e-mail consent
Enrolment welcome e-mail to parent	Works	Sent automatically at enrolment
Payment link e-mail to parent	Works	Sent from the payment screens
Staff-written e-mail about a student to parent	Not built	Saved on the student as "Recorded (not sent)"
Receiving e-mail from parents and clients	Not built	No incoming e-mail handler exists; only WhatsApp replies come back in
Parent replies	Needs fix	Mail goes from `no-reply@` with no Reply-To address, so replies are lost

Part 1: Outgoing e-mail (settings only)

Outgoing e-mail needs no development: set up Resend, add three DNS records and turn the integration on. Owner: CRM administrator plus whoever manages the domain's DNS.

1. **Create a Resend account** at resend.com and add the sending domain under **Domains**, for example `labofuture.com`.
2. **Add the DNS records Resend shows** (SPF, DKIM, DMARC) at the domain provider. Wait until the domain shows **Verified**.
3. **Create an API key** in Resend with sending access.
4. **Store the key as a Worker secret:**
   - Production: `npx wrangler secret put RESEND_API_KEY`
   - Local testing: add `RESEND_API_KEY=...` to `.dev.vars` and set `INTEGRATION_MODE=auto` (it is `mock` by default, which sends nothing real).
5. **Set the sender** in `wrangler.toml`: `EMAIL_FROM = "Lab of Future <admissions@labofuture.com>"`, then deploy.
6. **Turn it on in the CRM:** Configuration → Integrations → **E-mail**.
7. **Check consent:** leads without e-mail consent are marked "Blocked – no consent" and never sent. Capture consent on the lead form.
8. **Test:** Settings → Testing console → E-mail → **Test connection**, then send a template from a test lead and confirm it in the **Integration log**.

Part 2: Incoming e-mail (development needed)

Incoming e-mail should use Cloudflare Email Routing, because the CRM already runs on Cloudflare Workers. This requires the domain's DNS to be managed by Cloudflare.

Setup (administrator / IT)

1. Move the domain's DNS to Cloudflare, if it is not there already.
2. In Cloudflare, turn on **Email Routing** and create the addresses, for example `admissions@` and `info@`.
3. Route those addresses to the **lof-crm Worker**.

What the CRM will do (development)

1. Receive each message in a new e-mail handler in the Worker and read the sender, subject and body.
2. Match the sender's address to a **lead** or to a **parent on a student record**.
   - Match found: save it as an incoming e-mail on that record, add a timeline entry and alert the assigned counsellor.

    - No match (new client): create a new lead with source "E-mail", or place it in an "Unmatched e-mails" list for review.
3. Add a **Reply-To** of `admissions@` to every outgoing e-mail, so parent replies come back into the CRM.
4. Log every incoming message in the Integration log, with the same redaction as other integrations.
5. Add an "Incoming e-mail" simulation to the Testing console.

If the DNS cannot move to Cloudflare, an inbound e-mail service (Resend Inbound or Postmark) can post messages to a CRM webhook instead; the CRM side stays the same.

Part 3: E-mails about a student to the parent (development needed)

Staff will get a "Send e-mail to parent" action on the student screen, with the server deciding who may send and to which address.

1. **Templates:** progress update, attendance, fee reminder, general notice, plus free text. Managed under Configuration → Communication like the existing templates.
2. **Recipient:** the parent e-mail on the student record only; staff cannot type another address.
3. **Checks on the server:** the user's role allows it, the student is in the user's branch, and the parent has e-mail consent.
4. **Sending:** queued and sent through Resend with Reply-To `admissions@`.
5. **Record:** saved on the student's communication history as Sent or Failed, with the sender's name and time.
6. **Replies:** a parent's reply arrives through Part 2 and is attached to the same student.

Decisions and next steps

Decisions needed

- Is the domain's DNS on Cloudflare, or can it move there? This decides the incoming-mail route.
- Which addresses parents should write to (`admissions@`, `info@`, one per branch?).
- Unknown senders: create a new lead automatically, or hold them for review?
- Which roles may send e-mails about a student, and which templates they need.

Checklist

- [ ] Resend account created and domain verified
- [ ] `RESEND_API_KEY` secret and `EMAIL_FROM` set

- [ ] E-mail integration turned on and Test connection passed
- [ ] Domain DNS on Cloudflare and Email Routing addresses created
- [ ] Incoming e-mail handler built and tested (Part 2)
- [ ] "Send e-mail to parent" built and tested (Part 3)
- [ ] Reply-To added to outgoing e-mail
- [ ] End-to-end test: parent e-mails in, counsellor replies, reply lands on the student